

Thesis Project

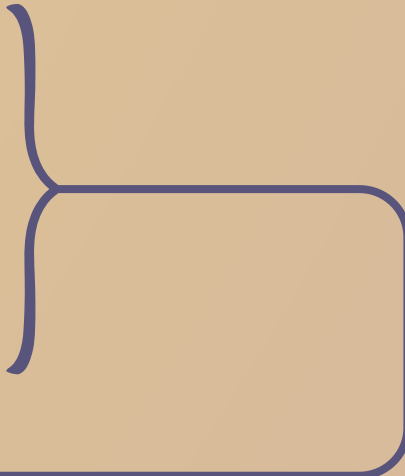
Estimation of Metabolic Fluxes with GNNs

Aim of the Project

The Problem and the Objective

Metabolic flux prediction is a notoriously complex problem.

Constraint-based techniques like **Flux Balance Analysis** (FBA) have several limitations:

- Lack of **experimental flux measurements**
 - Fixed **cellular objectives** & strict **steady-state** assumptions
 - Limited **multi-omics** integration
 - Execution **time**.
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These problems could be addressed with a **Neural Network** (NN) model.

Using a **Graph Neural Network** (GNN), the organization of metabolism becomes integral to the training.

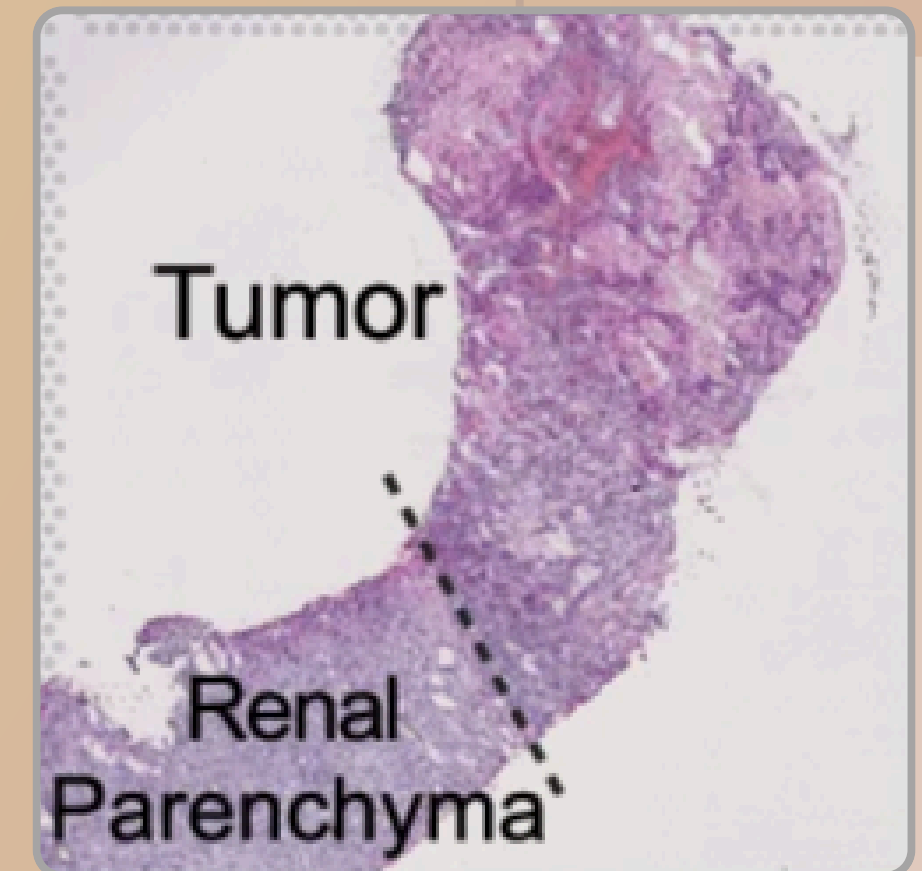
Preprocessing

Data Provenance and Preprocessing

Clear cell renal cell carcinoma (**ccRCC**) **spatial RNA-Seq** (Visium) sample 6800STDY12499506.
The data is already normalized and log-scaled.

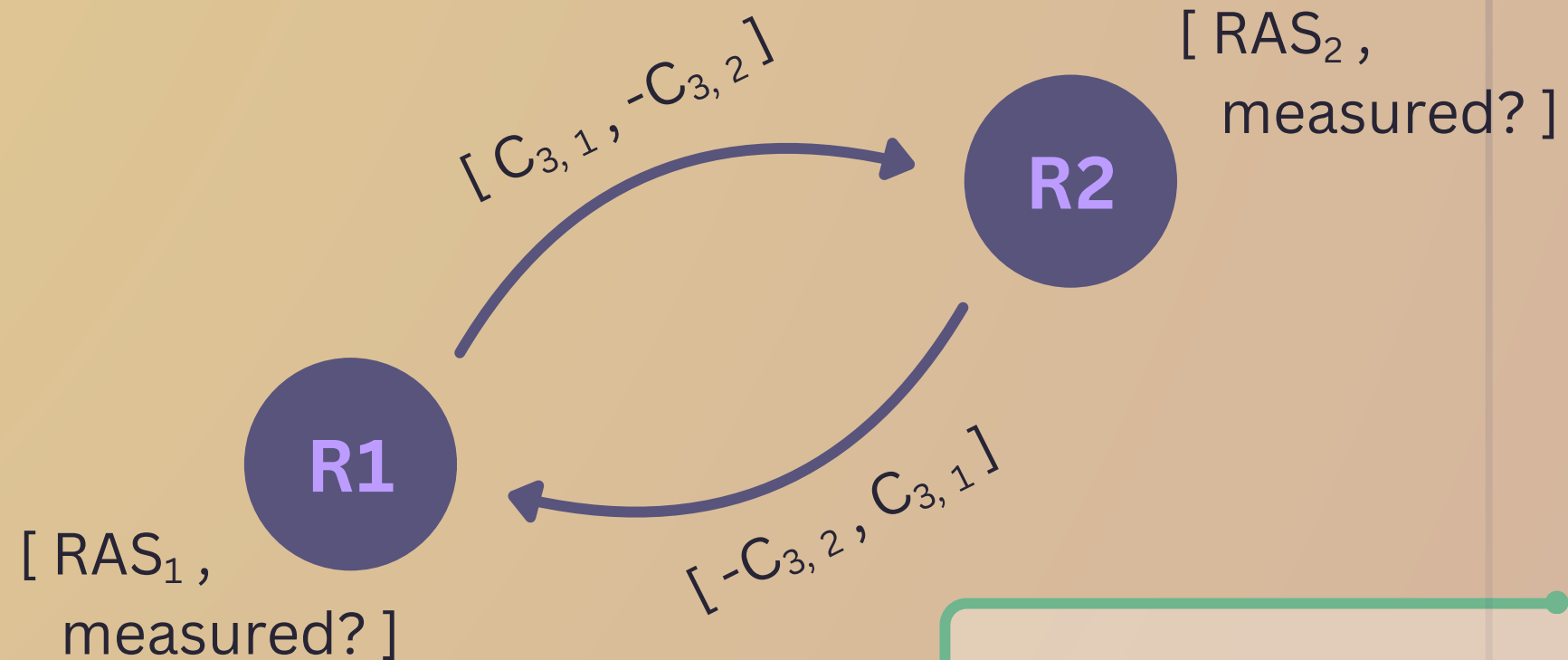
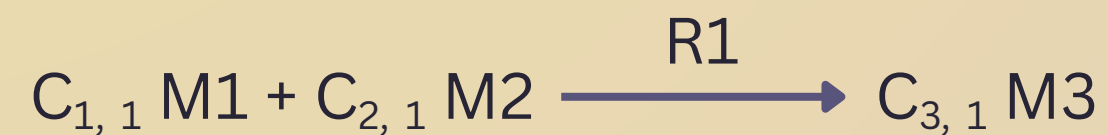
Max-normalization is then applied to the Reaction Activity Scores (**RAS**).

The **ENGRO2** metabolic model was used in all my tests so far..
removing the **boundary** reactions leads to **better mass balance**.



The Architecture

Metabolic Graphs and Message-Passing



Cofactors and similar **hub metabolites** were removed.

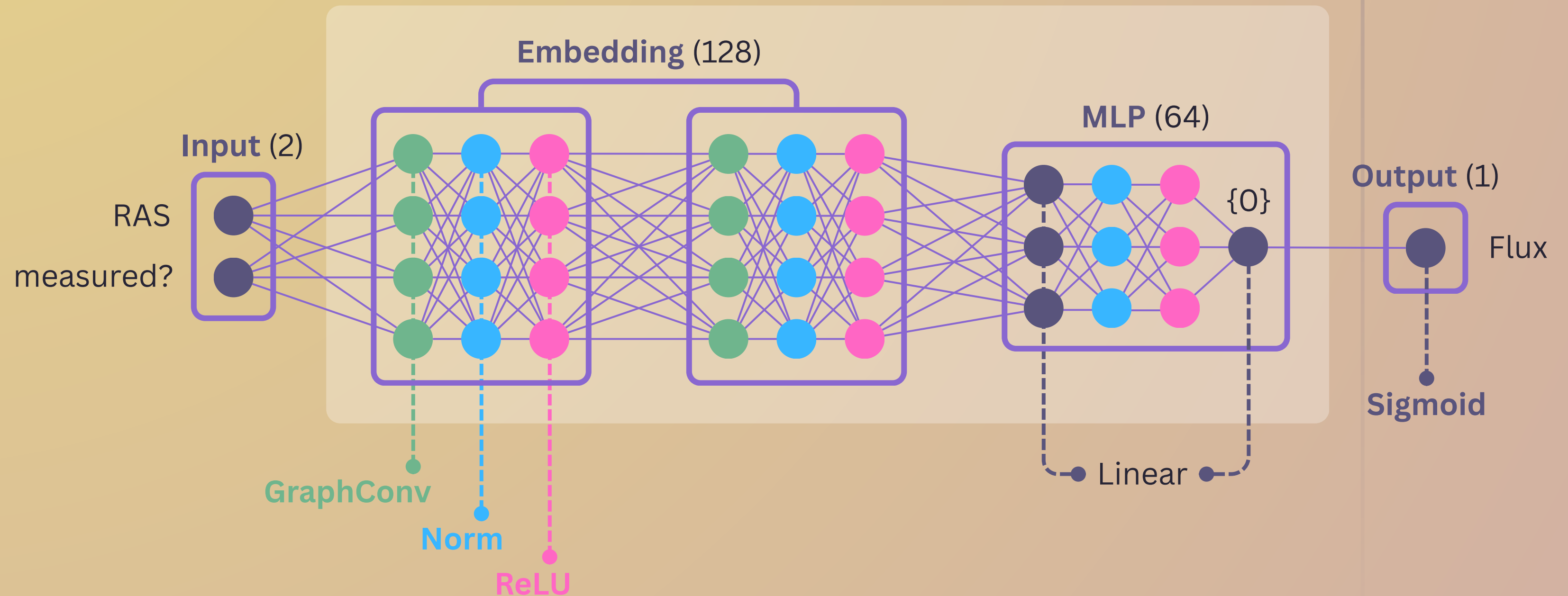
The double edge feature can teach the network the difference between **consumer-consumer** and **consumer-producer** relationships between reactions.

$$x'_i = W_1 x_i + W_2 \sum_{j \in N(i)} e_{j,i} x_j$$

GraphConv

The Architecture

Neural Network Layers



The Architecture

Loss Function

The model is currently trained on these **loss components**:

- **Mass balance**: $|SF^T|$
- Minimum **maintenance**: $ReLU(1\%FBA_m - |Biomass| + |ATP_m|)$
- **RAS** correlation: $1 - spearman(F, RAS)$
- **Pathway score** correlation: $1 - pearson(Biomass, score)$

Results

My Benchmarks

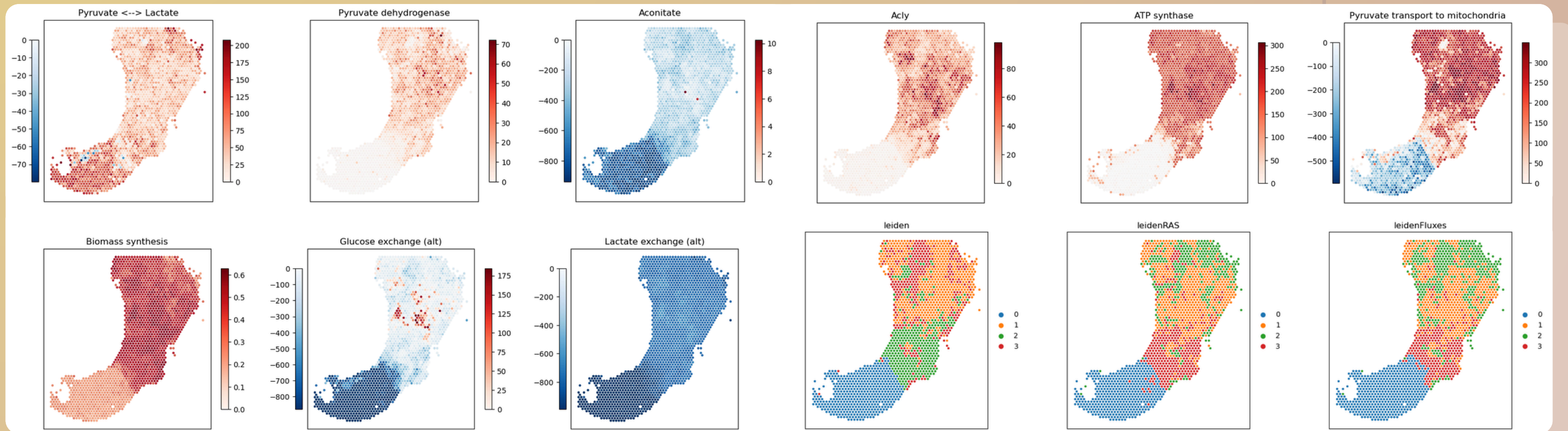
The quality of the predicted fluxes is evaluated in several ways:

- **Spatial plots** for specific reactions of interest
- **Clustering similarity** with the **RAS** and the original data
- **Proliferation score** correlation
- Clustering similarity with **GSEA metabolic pathway** scores (KEGG 2021 Human)
- Comparison with **flux sampling** results
- Distribution of **mass imbalance**

Results

Current Performance

Training the model for **500 epochs** gives decent results, when using MAGIC:



Silhouette score: 0.24
V-measure vs fluxes: **0.62**

Silhouette score: 0.27
V-measure vs fluxes: **0.80**

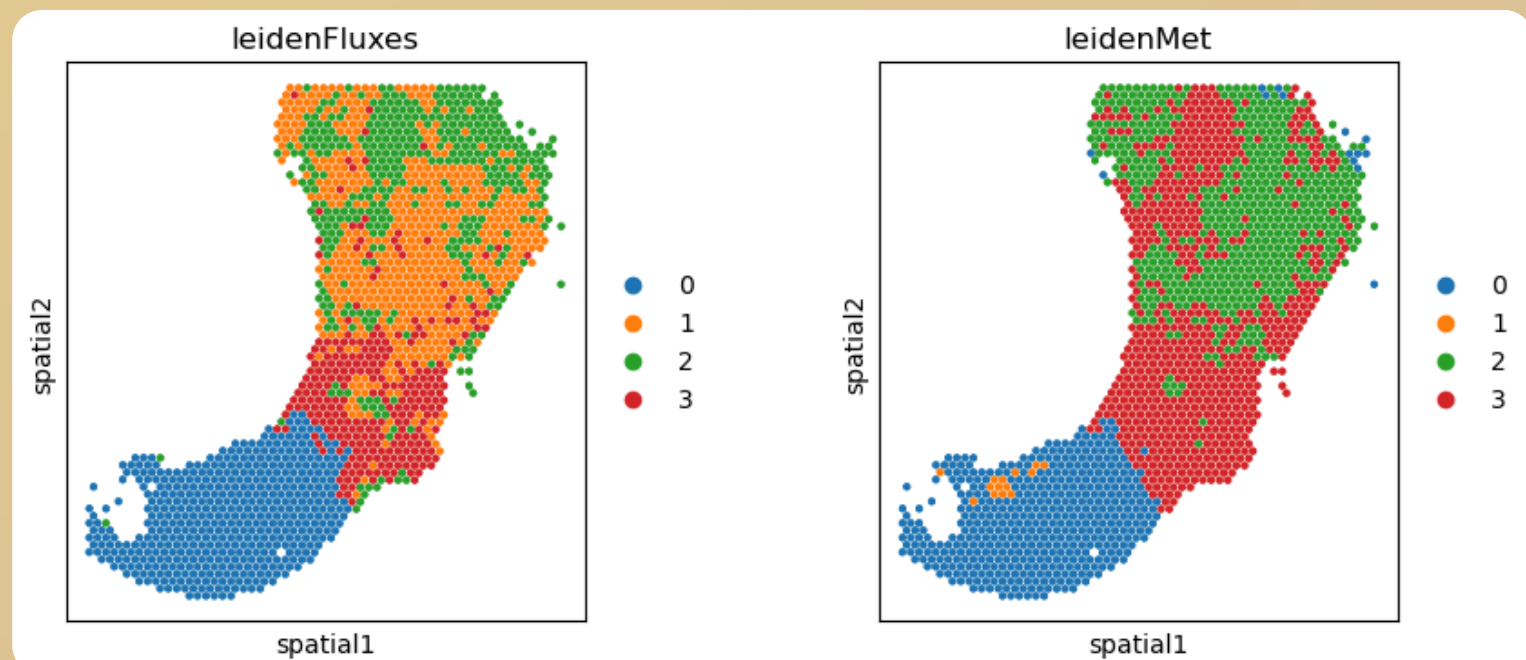
Silhouette score: 0.23

Results

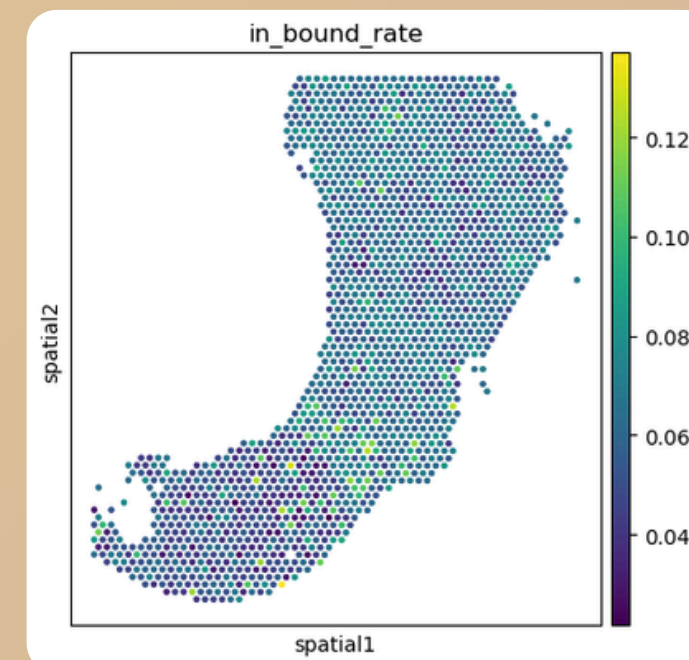
Current Performance

The **Biomass flux** of each spot significantly correlates with its **proliferation score** by **94.97%** (Spearman), **97.83%** (Pearson).

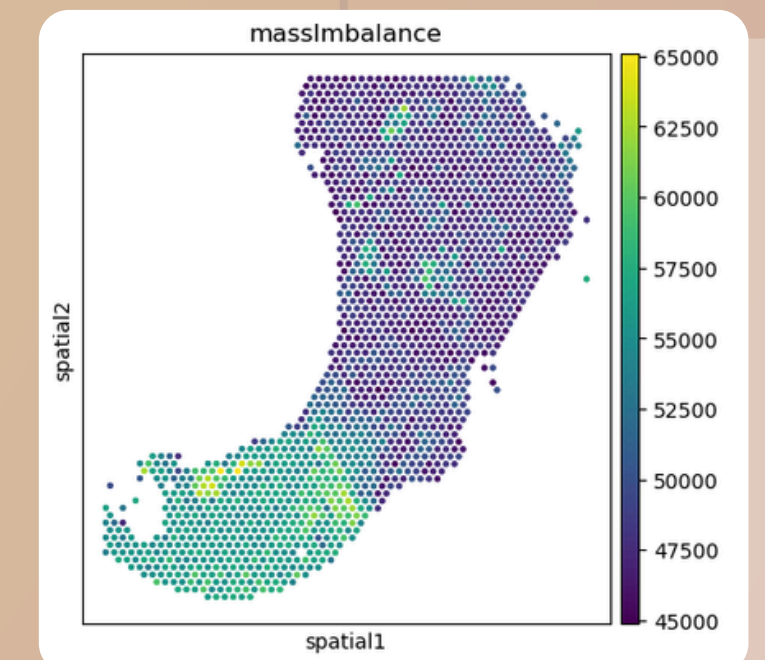
The predicted fluxes significantly correlate with **CBS normalized mean fluxes** by 51.74% (Spearman).



V-measure fluxes vs pathway scores: 0.53



Average in-bound rate: 0.061



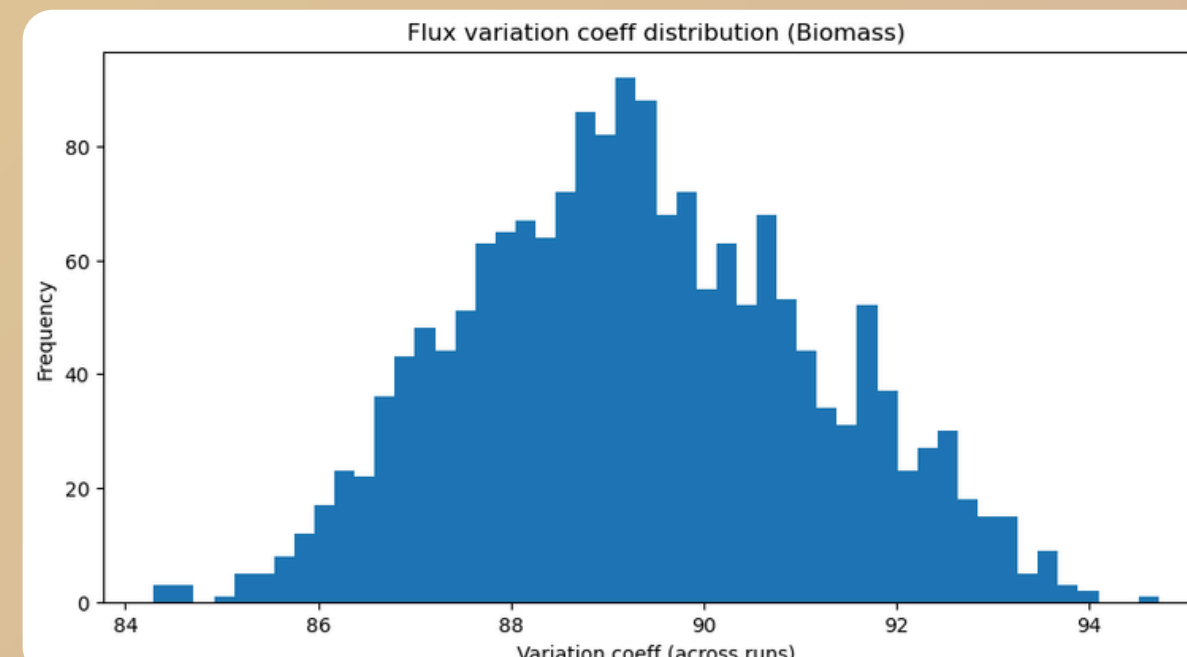
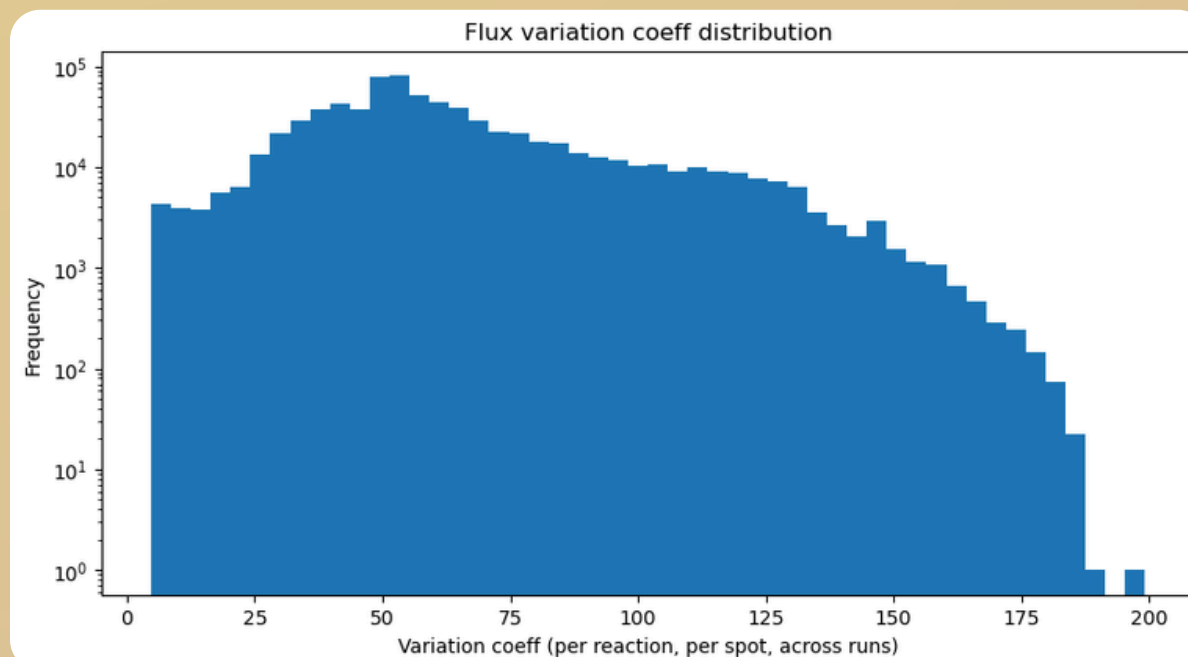
Results

Latest Tests

After the latest modifications to the tool, I ran these tests:

- **Other curves** in the last GNN layer: tanh, modified tanh (smht) & softsign
- Modified ENGRO2: removed **palmitate** and **galactose** from the medium
- **Random seed** variance test:

} No appreciable differences.



Future Developments

What I Plan to Add Next

Aside from solving the more urgent problems, my plans for upcoming features include:

- Re-introducing **hub metabolites** in graph construction
- Reproducing expected fluxes in **hypoxia** datasets
- **Multi-omics** integration: and support for non-spatial data
- **Thermodynamical constraints** as an additional loss component
- Output of **flux distributions** instead of point predictions